REDUCEDNESS CHECKED ON ALL RINGS OF SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2376-2386, following the statement at 2370-2374.

Allowed standard prerequisites (editorial).

- Sections vanishing on all stalks are zero. Sheaves, Lemma sheaves-lemma-sheaf-subset-stalks (cross-chapter, cited at ~2381). Forward direction conclusion.

Source reference. Original reference 0079: focused-seeds/geometry/S9-schemes-reduced/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *A scheme X is reduced if and only if $\mathcal{O}_X(U)$ is a reduced ring for all $U \subset X$ open.*

Proof. Assume that X is reduced. Let $f \in \mathcal{O}_X(U)$ be a section such that $f^n = 0$. Then the image of f in $\mathcal{O}_{U,u}$ is zero for all $u \in U$. Hence f is zero, see Sheaves, Lemma 0079. Conversely, assume that $\mathcal{O}_X(U)$ is reduced for all opens U. Pick any nonzero element $f \in \mathcal{O}_{X,x}$. Any representative $(U, f \in \mathcal{O}(U))$ of f is nonzero and hence not nilpotent. Hence f is not nilpotent in $\mathcal{O}_{X,x}$. □

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.